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Michael Herman

AI Researcher / Data Scientist

🔍 Google Scholar ORCID
🐙 GitHub Hugging Face
📄 Zenodo Tableau Public
🌐 LinkedIn Portfolio

AI and computer vision researcher with 5+ years of experience developing ML models for satellite-based sensor systems. Specialized in deep learning and state estimation algorithms. Demonstrated success publishing novel space sensor DL models, large-scale image datasets, and a comprehensive sensor algorithm survey. Passionate about advancing AI for defense applications.

EDUCATION

- Georgia Institute of Technology** Atlanta, GA
Ph.D. in Aerospace Engineering; GPA: 3.58 Jan 2018 - Dec 2024
- Georgia Institute of Technology** Atlanta, GA
M.S. in Aerospace Engineering; GPA: 3.58 Jan 2015 - Dec 2017
- Georgia Institute of Technology** Atlanta, GA
B.S. in Aerospace Engineering; GPA: 3.61 Aug 2012 - Dec 2014
Honors: Highest Honors

SKILLS SUMMARY

- Languages** Python, MATLAB, R, JAVA, C, FORTRAN, PowerShell
- Frameworks** PyTorch, TensorFlow
- Libraries** iNNvestigate, Matplotlib, NumPy, OpenCV, Pandas, Scikit-learn, SciPy, Seaborn, SpConv, Tensorboard, Torchvision
- Tools** Anaconda, Docker, GIT, Jupyter, LaTeX
- Software** NI LabVIEW, Ansys Zemax OpticStudio, MATLAB, Simulink, STK, Tableau, Unreal Engine 5

PROFESSIONAL EXPERIENCE

- Georgia Institute of Technology** Atlanta, GA
IT Support Engineer Senior - College of Engineering Feb 2025 - Current
 - Worked with the Georgia Tech AI Makerspace to assist users in developing Artificial Intelligence (AI) tools and managing High-Performance Computing (HPC) GPU clusters for the Partnership for an Advanced Computing Environment (PACE).
 - Developed custom scripts to support a modern SCCM-based Windows imaging workflow, replacing the legacy MDT workflow across campus.
 - Collaborated with IT staff and researchers across the College of Engineering units to provide hardware and software technical support.
- Georgia Institute of Technology** Atlanta, GA
Graduate Teaching Assistant - Aerospace Engineering IT Support Aug 2019 - Dec 2024
 - Provided on-site support for students, faculty and staff in the Aerospace Engineering department to troubleshoot software and hardware.
- Georgia Institute of Technology** Atlanta, GA
Graduate Teaching Assistant - Space Systems Design Laboratory (SSDL) Jan 2015 - Dec 2015
 - Taught, graded and made assignments for the senior design class of the undergraduate space systems track.

RESEARCH EXPERIENCE

- Georgia Institute of Technology** Atlanta, GA
Graduate Research Assistant - Aerospace Systems Design Laboratory (ASDL) Aug 2019 - Dec 2024
 - Prototyped and published sparse submanifold convolutional neural network (SSCNN) for sun sensor attitude determination to HuggingFace.
 - Applied data fusion to a Multiple-Input Multiple-Output (MIMO) deep neural network (DNN) for multi-sensor input and two-axis state output.
 - Developed a physics-based diffraction simulation for model training and testing using OpticStudio and MATLAB via ZOS-API to Github.
 - Created and publicly released a large-scale labeled image dataset (2M+ images) for digital sun sensor model training, hosted on Zenodo/HF.
 - Quantified and analyzed the performance and robustness of the algorithm via error histograms and heatmaps using the Seaborn library.
 - Estimated uncertainty of the deep learning model with Bayesian inference using Monte Carlo Dropout (MCDropout).
 - Applied saliency mapping to assess AI explainability of the output image space using the iNNvestigate library.
- Georgia Institute of Technology** Atlanta, GA
Graduate Research Assistant - Space Systems Design Laboratory (SSDL) Jan 2016 - Dec 2017
 - Developed and documented algorithms for an analog sun sensor in the Ranging And Nanosatellite Guidance Experiment (RANGE) mission.
 - Designed an ADCS CubeSat test platform with spherical air bearing test-bed incorporated within a Helmholtz cage for in-house testing.
 - Implemented an Extended Kalman Filter (EKF) within the CubeSat flight control system to estimate and correct analog sun sensor biases.

PROJECTS

- Dissertation:** Herman, M. (2025). Predictive calibration for digital sun sensors using sparse submanifold convolutional neural networks.
- Masters Project:** Herman, M. (2016). Design and Application of a Circular Aperture Sun Sensor.

SELECTED PUBLICATIONS

- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Sun sensor calibration algorithms: A systematic mapping and survey [under-review].
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Sun sensor calibration algorithms: Survey dataset (1.0) [Data set]. Zenodo.
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). Predictive calibration for digital sun sensors using sparse submanifold convolutional neural networks [under-review].
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). DSS-CNN: calibration image dataset (1.0) [Data set]. Zenodo.
- Herman, M., Pinon Fischer, O., & Mavris, D. N. (2025). DSS-CNN: calibration models (1.0) [Model]. Zenodo.